Electromagnetic interference [EMI]

Electromagnetic interference is the degradation in the performance of a device due to the fields which make the electromagnetic environment.

Electromagnetic environment in the apparatus include radio, TV, Broadcast station, radar etc....

Every electronic device is susceptible to EMI. ie) every electronic device suffer from EMI difficulties.

There are two types of EMI.

① Intra EMI

② Inter EMI

① Intra EMI

If the functional characteristics of one module with in a system is disturbed due to EMI from another module.

② Inter EMI

The functionalities of one equipment are disturbed due to other equipment.

Sources

EMI can be classified in terms of its causes and sources. classification will help recognition of sources and assist in determining means of the control.

sources of EMI are classified into two types namely natural or artificial.

① Natural sources

Natural are of terrestrial and extra terrestrial which include

- ↳ atmospheric thunderstorms
- ↳ lightning discharge
- ↳ cosmic noise
- ↳ radio stars

② Man made or Artificial sources

(i) Electric power sources

- ↳ Generators, faults transformers, faults insulators
- ↳ Transmission equipments

↳ Distribution equipments like Faulty wiring, poor grounding etc...

(ii) Electronic communications devices

↳ Police radio

↳ TV and radio

↳ point to point communication

↳ satellite communication

↳ Radar

③ Machines and tools

↳ Industrial machines

↳ Office or business machines

↳ welders, heaters, transformers, power tools

↳ Appliances such as AC, Fridge, ovens, vaccum cleaners etc...

④ Ignition systems

↳ Engines

↳ Vehicles such as tanks, trucks, tractors, automobiles, aircrafts etc...

↳ Tools

⑤ consumer and medical electric systems.

↳ Lights such as fluorescent lamps, neon lamps etc...

↳ Entertainments such as home PC, recorder, TV recievers

↳ Appliances

↳ Medical equipments such as ultrasonic scanners

⑥ Nuclear things such as nuclear aircraft, ships, nuclear power stations etc...

Characteristics of EMI

The origin of EMI are basically conducted emission OR radiated emission.

Conducted emissions

These are those currents [voltage] that are carried by metallic paths and placed on the common power network where they may cause interference with other devices that are connected to this network.

Radiated emissions

These are concerned with electric or magnetic fields that are radiated by the device that may be recieved by another device. This will result in interference among those devices.

Common Effects of EMI

- Annoying effects
- Disturbing effects
- Effects like burning of equipments, loss of data.

Electro magnetic compatibility [EMC]

Electro magnetic compatibility is defined as the ability of the equipment to operate satisfactorily in the presence of interference and not be the source of interference for the neighbour.

EMI and EMC are related in such a manner that EMI is the problem that occurs when unwanted voltage or current are present to influence the device performance. and EMC is the solution to that problem.

Goal of EMC is to make sure [ensure] that system o. is compatible and this is achieved by applying proper design techniques the use of which ensures a system relatively free from EMI problems.

EMI Control Techniques

Techniques used to suppress or control EMI are

① Grounding

② shielding

③ Filtering

① grounding

↳ It is the establishment of an electrically conductive path between two points to connect electrical and electronic elements of a system called ground.

↳ Floating ground is used to isolate circuits and equipments electrically from a common ground plane

Two types of grounding

↳ single point grounding

↳ Multiple point grounding

② Shielding

Shielding is used to radiate energy to a specific region to prevent radiated energy from entering a specific region.

$$\text{Shield effectiveness} = 10 \log_{10} \frac{\text{Incident power density}}{\text{Transmitted power density}}$$

Incident power density → It is the power density at a point before a shield is installed.

Transmitted power density → It is the power density at the same point after the shield is in place.

(3) <u>Filtering</u>

It is a network of lumped or distributed components such as resistors, inductors and capacitors that offers comparatively little opposition to certain frequencies, while ~~opposition~~ blocking the passage of other frequencies.

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